METRIC

MIL-PRF-8565/7A
w/AMENDMENT 3
13 January 2022
SUPERSEDING
MIL-PRF-8565/7A (AS)
w/AMENDMENT 2
15 September 2017

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MAINTENANCE-FREE, 24-VOLT, 24-AMPERE-HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.

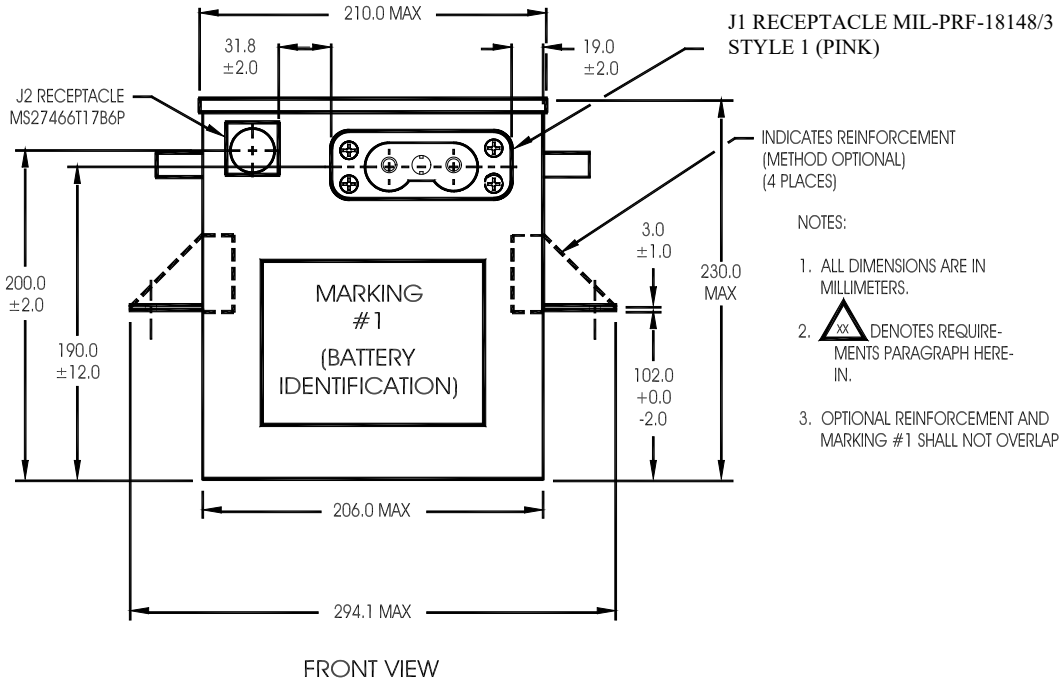
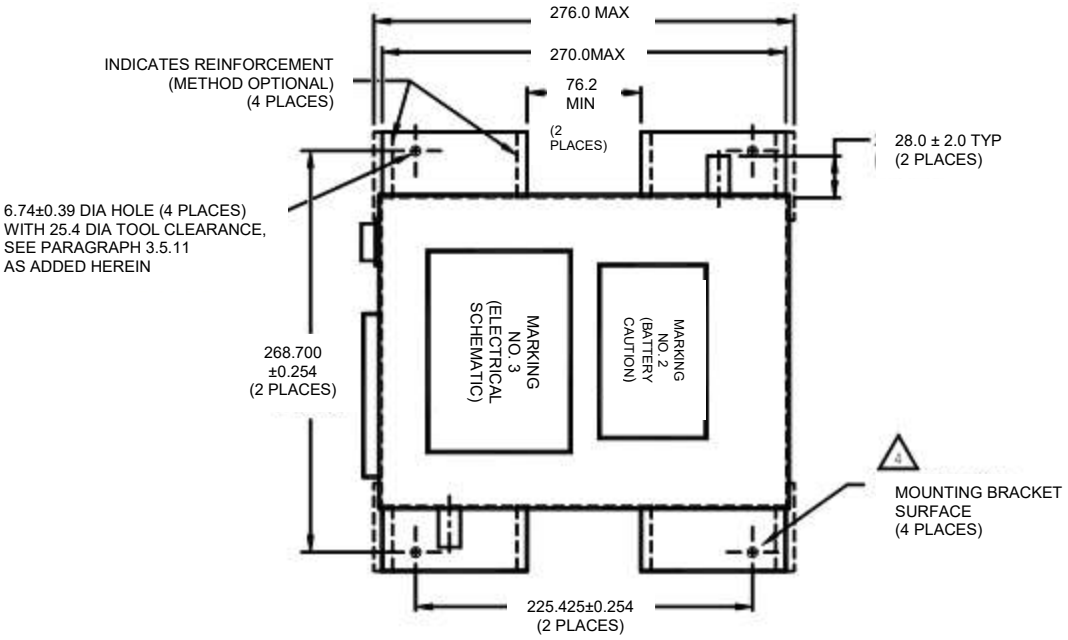
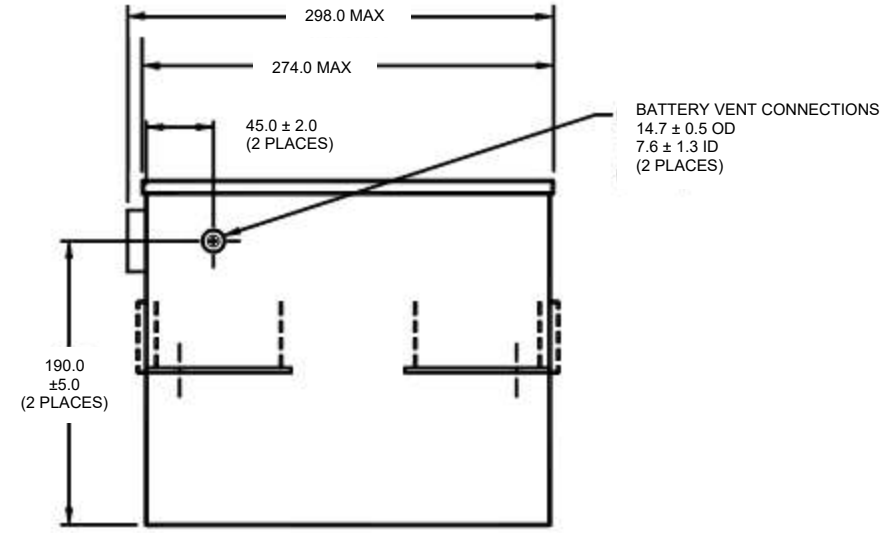


FIGURE 1. Dimensions and configuration for PIN D8565/7-2.

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TOP VIEW



SIDE VIEW

FIGURE 1. Dimension and configuration for PIN D8565/7-2 – Continued.

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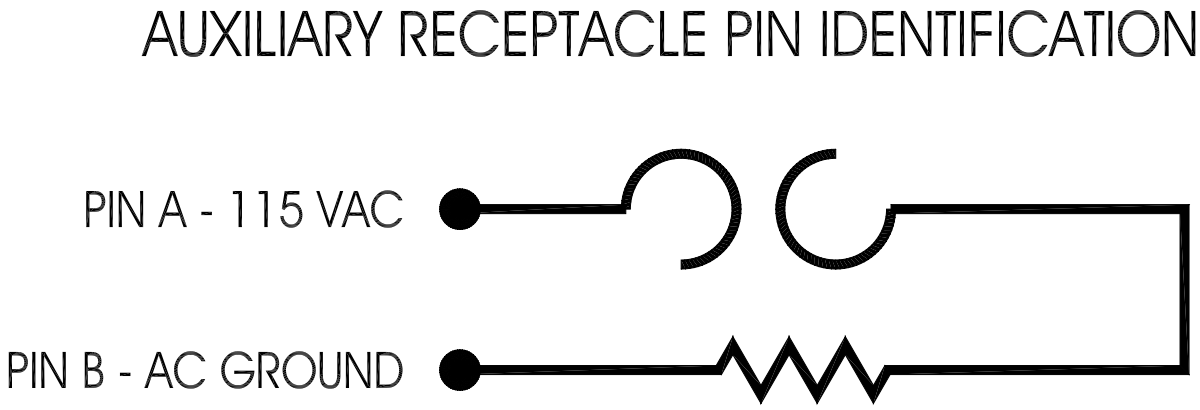
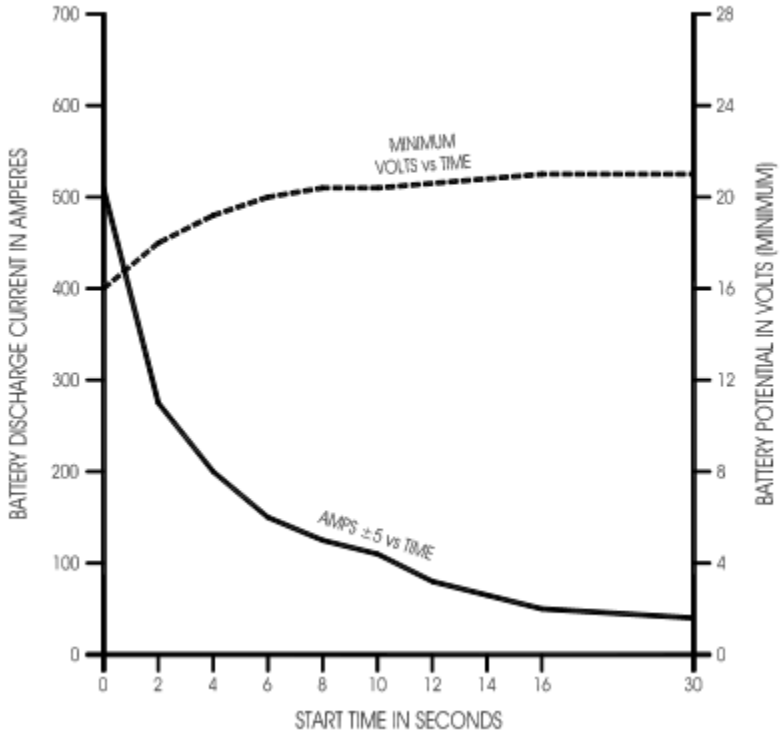


FIGURE 2. Electrical schematic for marking #3.

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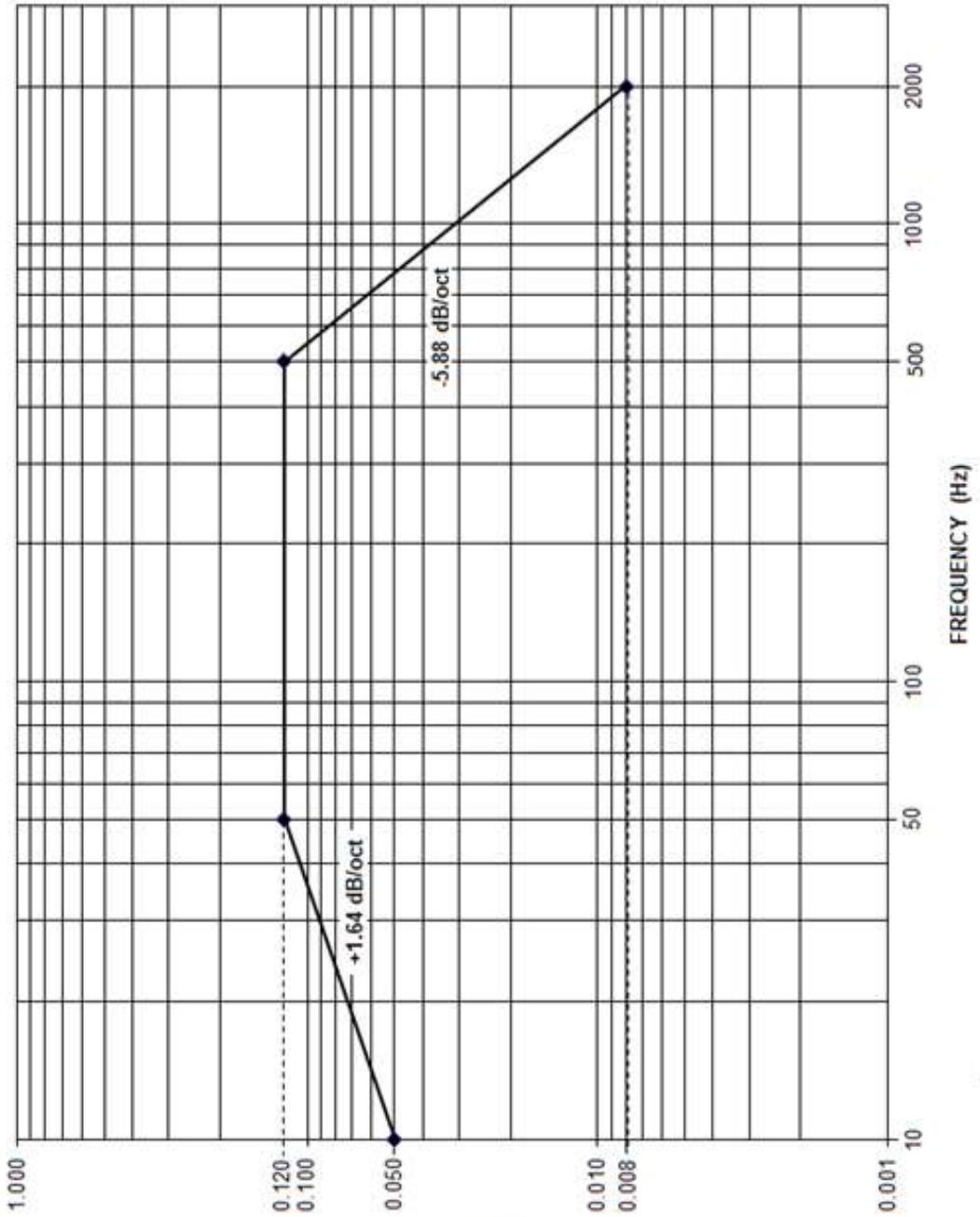
COORDINATES OF CURVES STRAIGHT LINE BETWEEN COORDINATES		
CURRENT IN AMPERES	POTENTIAL IN VOLTS	START TIME IN SECONDS
510	16.0	0
275	18.0	2
200	19.0	4
150	20.0	6
125	21.0	8
110	21.0	10
80	21.5	12
65	22.0	14
50	22.5	16
40	22.5	30



NOTE:
1. The abbreviations used on this figure are amperes (AMPS) and verses (vs).

FIGURE 3. Start cycle requirements.

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POWER SPECTRAL DENSITY
g^2/Hz

NOTE:
1. Abbreviations used on this figure are acceleration of gravity (g), decibel (dB), Hertz (Hz), and octave (OCT).

FIGURE 4. Random vibration test levels.

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REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery of this specification sheet shall be D8565/7-2. The D8565/7-2 battery supersedes the D8565/7-1 battery.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1 contained in the specification sheet. The weight of the battery shall be not greater than 29.0 kilograms (see 3.8.3 and 4.5.3). In addition, the weight of any production battery shall vary by not greater than ±0.6 kg from the average weight of the three approved qualification sample batteries but may not exceed the 29.0-kilogram limit.

3. Capacity. The rated capacity of the battery shall be not less than 24 ampere-hours (Ah). See 3.8.6 as modified herein.

4. Battery mounting. The battery's side mounting brackets shall be used to support the battery as shown on figure 3 of the general specification. If vibration isolators are used, sufficient clearance between the battery and any structure shall be provided to ensure that the battery will not strike the structure under any swaying motion.

5. Service life. The battery's service life (useful life) in aircraft shall be not less than 3 years. The battery shall require no scheduled maintenance during the 3 years (see 3.8.9, 4.5.10, and 6.11.7).

6. Qualification inspection and conformance inspection. Table I lists the required inspections.

7. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

7.1 Change "18.0 volts" to "20.0 volts" in the following paragraphs:
- 4.5.6 Electrical test preparation, first sentence.
- 4.5.7 Capacity discharge, second sentence.
- 4.5.11 Capacity discharge at 49 °C (120 °F), third sentence.
- 4.5.17 Temperature rise and float test, step e, second sentence.
- 4.5.23 Physical integrity at high temperature test, steps b and k.
- TABLE III Capacity performance requirements, requirements (1), (2), and (3).

7.2 Modify the following paragraphs:

3.5.5 Receptacles. Insert the following after the first sentence: "Each battery shall include the following receptacle(s) as its electrical interface to the aircraft. The J1 receptacle (main receptacle) shall be MIL-PRF-18148/3, style 1 (pink). It shall contain the battery's positive and negative electrical power connections and shall be mounted as shown on figure 1 of the specification sheet. The J2 receptacle (auxiliary receptacle) is required only when heaters are incorporated into the battery (see 3.5.10). The receptacle shall be equivalent to MS27466T17B6P and mounted as shown on figure 1 of the specification sheet. It shall contain the electrical connections for the AC power to the battery's internal heaters (see figure 2 of the

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specification sheet). Pin A shall be AC positive and pin B shall be AC negative (ground)."

3.5.10 Heaters. In 3.5.10.a, insert the following after the last sentence: "Heaters shall be powered from the battery's auxiliary receptacle pins A and B." In 3.5.10.b, delete the text and insert the following: "Heater current shall be not greater than 3.0 amperes at 115.0 VAC. The rated current of each heater control device shall be compatible with the heater current." In 3.5.10.h, delete "in accordance with any specified environmental condition" and insert "at 24 °C".

7.3 Add the following paragraph:

"3.5.11 Mounting holes. The mounting bracket surfaces that surround each mounting hole shall be free of any coating for a distance not less than 12.7 millimeters radius from the center of the hole. This shall apply to both the top and bottom mounting surfaces."

7.4 Modify the following paragraphs:

3.6.5 Marking for batteries with heaters. At the end of the third sentence, insert: "(see figure 2 of the specification sheet)."

3.8.6 Capacity and electrical performance. Delete the text and insert: "The rated capacity of the battery shall be not less than 24.0 ampere-hours. To meet the capacity requirement, each battery shall produce a potential of not less than 20.0 volts at not less than 60 minutes when connected to a 24.0-ampere load after being tested in accordance with 4.5.7 at 24 °C (75 °F) (24.0 Ah/1.0 hr./24 °C/20.0 V at end of life). The battery's 1C-rate shall be 24.0 amperes (see requirement 3 of the specification sheet)."

3.8.7 Constant voltage discharge. Delete the first sentence and insert: "The battery shall produce not less than the current specified below after being tested in accordance with 4.5.8."

Temperature	Current at peak value (Time =0)	Current at 15 seconds	Current at 30 seconds	Capacity at 60 seconds
Room temperature (see 4.4.1)	1150 amperes	875 amperes	725 amperes	12.7 ampere-hours
-26.0 ±2.0 °C	725 amperes	575 amperes	400 amperes	6.6 ampere-hours

3.8.13.1 Ground storage. Add after the first sentence: "After the storage period, the battery shall meet the requirements of 3.8.18 after being tested in accordance with 4.5.22."

3.8.16 Output performance. Delete the text and insert: "The battery shall produce a potential of not less than 24.0 volts during and after the test of 4.5.28. The battery shall also meet the requirements of 3.8.18 for not less than three successive cycles."

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3.8.17 Cold and hot temperature charge. Delete the first sentence and insert: "The battery shall produce not less than 12.0 ampere-hours after being tested in accordance with step 4.5.29.1.c. The battery shall produce not less than 18.0 ampere-hours in not less than 45 minutes after being tested in accordance with steps 4.5.29.1.h and 4.5.29.2.e."

7.5 Add the following paragraphs:

"3.8.18 Start cycle. The battery shall produce not less than the specified amount of current and voltage within the specified time period shown on figure 3 of the specification sheet after being tested in accordance with 4.5.22 and 4.5.28."

7.6 Modify the following paragraphs:

4.5.8 Constant voltage discharge. For group B conformance inspection: Delete steps 4.5.8.g and 4.5.8.h.

4.5.12 Charge and discharge test at low temperature. Not applicable.

4.5.15 Shock test (basic design). Delete steps 4.5.15.e and 4.5.15.f and insert the following:

"e. Perform the following between each of the shocks of step 4.5.15.d: Dismount the battery from the shock test apparatus. Examine the battery for the requirements of 3.7. Repeat steps 4.5.15.b and 4.5.15.c.

"f. After the fifteenth impact, examine the battery for the requirements of 3.7. Constant potential charge the battery in accordance with 4.4.2. Discharge the battery at the 6.5C-rate for 3 minutes."

4.5.19 Vibration test. Delete steps 4.5.19.a through 4.5.19.h and insert the following:

"a. Constant potential charge the battery in accordance with 4.4.2.

"b. Connect two AN-6 cables to the positive terminal and another two AN-6 cables to the negative terminal. Clamp the cables to the test fixture at a cable length of 46 ±3 centimeters from the battery.

"c. Mount the battery assembly, including connector and cables, to the test fixture via the normal battery attachment points (see 4.4.5, requirement 4 of the specification sheet, and figure 3 of the general specification).

"d. Discharge the battery at the 6.5C-rate for 1 minute. Examine the battery for the requirements of 3.8.10.

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"e. Constant potential charge the battery in accordance with 4.4.2 during vibration. Vibrate the battery along each orthogonal axis (in the axis order shown on figure 3 of the general specification) in accordance with random vibration levels on figure 4 of the specification sheet. Vibrate each axis for 30 minutes.

"f. Perform the following between each axis of vibration during step 4.5.19.e: Dismount the battery from the vibration apparatus. Examine the battery for the requirements of 3.7. Repeat steps 4.5.19.b through 4.5.19.d.

"g. After completion of the vibration test, dismount the battery from the vibration apparatus. Examine the battery for the requirements of 3.7. Constant potential charge the battery in accordance with 4.4.2.

"h. Discharge the battery at the 6.5C-rate for 3 minutes.

"i. Examine the battery for the requirements of 3.7 and 3.8.10."

4.5.22 Ground storage test. Delete the text of step 4.5.22.d and step 4.5.22.i and insert into both steps: "Discharge the battery in accordance with figure 3 of the specification sheet for three successive cycles. The battery shall rest on open circuit for 2.0 ±0.1 minutes between the end of one cycle and the start of another cycle." Insert at the end of step 4.5.22.e and step 4.5.22.j: "and 3.8.18."

4.5.28 Output performance test. Delete the title and text.

7.7 Add the following paragraphs:

"4.5.28 Output performance test (standby power output). This test consists of the following steps.

"a. Constant potential charge the battery in accordance with 4.4.2. Open circuit the battery.

"b. Stabilize the battery's temperature at the test temperature specified below for not less than 12.0 hours. With each battery still in the temperature chamber at the indicated test temperature, discharge each battery into a 16-ohm load as follows:

Battery sample number	Test temperature	Discharge time
2	-26°C (-15°F)	1.0 hour
3	-18°C (0°F)	2.0 hours
1	0°C (32°F)	4.0 hours
3	25°C (77°F)	7.0 hours
1	49°C (120°F)	7.0 hours

"c. Examine the battery for the requirements of 3.8.16.

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"d. Remove the battery from the test temperature environment. Immediately discharge the battery in accordance with the discharge currents and discharge times (start cycle) shown on figure 3 of the specification sheet. Repeat the start cycle twice. Open circuit the battery for 2.0 ±0.1 minutes after completing a start cycle and before starting another start cycle.

"e. Examine the battery for the requirements of 3.8.16 and 3.8.18."

7.8 Modify the following paragraph:

4.5.29 Cold and hot temperature charge test. Delete the text.

7.9 Add the following paragraphs:

"4.5.29.1 Half-hour charge test. This test consists of the following steps.

"a. Constant potential charge the battery in accordance with 4.4.2.

"b. Open circuit the battery. Stabilize the battery (see 4.4.1) at -18°C (0°F).

"c. Keep the battery at the stabilized temperature of step 4.5.29.1.b while discharging at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"d. Examine the battery for the requirements of 3.7 and 3.8.17.

"e. Open circuit the battery. Stabilize the battery at 25 °C (77 °F) (see 4.4.1). Keep the battery at 25 °C (77 °F) while constant potential charging the battery in accordance with 4.4.2.

"f. Repeat step 4.5.29.1.b. Keep the battery at the stabilized temperature of step 4.5.29.1.b while discharging it at the 1.0 C-rate to remove all but 12.0 ampere-hours as determined by step 4.5.29.1.d.

"g. Repeat step 4.5.29.1.b. Constant potential charge the battery for 30 ±1.5 minutes in accordance with 4.4.2.

"h. Immediately after completing step 4.5.29.1.g, place the battery in a 25 °C (77 °F) environment. Within 5 minutes, discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"i. Repeat steps 4.5.29.1.a through 4.5.29.1.h above but change the temperature of step 4.5.29.1.b from –18 °C to 55 °C (0°F to 131°F).

"j. Examine the battery for the requirements of 3.8.17.

"4.5.29.2 One-hour charge test. This test consists of the following steps.
"a. Constant potential charge the battery in accordance with 4.4.2.

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"b. Open circuit the battery. Stabilize the battery (see 4.4.1) at -40 °C (-40 °F). Perform steps 4.5.29.2.c and 4.5.29.1.d with the battery in the environmental chamber at -40 °C (-40 °F).

"c. Discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"d. Repeat step 4.5.29.2.b. Constant potential charge the battery in accordance with 4.4.2 for 1.0 ±0.05 hour.

"e. Immediately after completing step 4.5.29.2.d, place the battery in a 25 °C (77 °F) environment. Within 5 minutes, discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"f. Examine the battery for the requirements of 3.7 and 3.8.17.

"4.5.30 Wormgear retaining pin hardness. Not applicable.

"6.11.7 Additional application information. The following information is provided for users of the D8565/7-2 battery.

"6.11.7.1 Initial installation date. The initial installer will write the date the battery is first installed in an aircraft (date first placed in service) in the space provided on the battery's identification marking (see 3.6.3 and figure 1 of the specification sheet).

"6.11.7.2 Service life. The battery's service life (useful life) in aircraft is limited to 3 years. A battery will not be used in aircraft after 3 years from the date first installed in an aircraft (see requirement 5 of the specification sheet).

"6.11.7.3 Removal date from aircraft. The battery will be removed from aircraft service 3 years after the date it was first installed in an aircraft.

"6.11.7.4 Maintenance-free. The battery will require no periodically scheduled maintenance during its service life (see requirement 5 of the specification sheet and 6.15.10)."

7.10 Modify the following figure:

FIGURE 1. Battery identification marking. Insert "Manufactured date and" in front of "Lot code".

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TABLE I. Qualification inspection and conformance inspection of batteries.

Test number	Examinations and tests 1/	Qualification inspection					Conformance inspection		
		Sample number			Requirement Paragraph 5/	Test Paragraph 5/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
1	INCOMING INSPECTION	X	X	X	3.8.1 (G)	4.5.1 (G)		X	X
2	VISUAL AND MECHANICAL EXAMINATION	X	X	X	Reqmt 4 (N), 3.5(S), 3.6 (S), 3.8.2 (G), table II (G)	4.5.2 (G)	X	X	X
3	DIMENSIONS AND WEIGHT	X	X	X	Reqmt 2(N), 3.8.3 (G)	4.5.3 (G)	X 2/	X	
4	DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST	X	X	X	3.7 (G), 3.8.4 (G)	4.5.4 (G)	X 2/	X	
5	STRENGTH OF VENT TUBES TEST	X	X	X	3.7 (G), 3.8.5 (G)	4.5.5 (G)	X	X	X
6	ELECTRICAL TEST PREPARATION	X	X	X	---	4.5.6 (S)	X	X	X
7	CAPACITY DISCHARGE	X	X	X	Reqmt 3 (N), 3.8.6 (S)	4.5.7 (S)		X	X
8	CONSTANT VOLTAGE DISCHARGE	X	X		3.7 (G), 3.8.7 (S), reqmt 4 of table III (G)	4.5.8 (S)	X	X	
9	STRENGTH OF RECEPTACLE TEST			X	3.7 (G), 3.8.8 (G)	4.5.9 (G)			
10	OUTPUT PERFORMANCE TEST (STANDBY POWER OUTPUT)	X	X	X	3.8.16 (S), 3.8.18 (N)	4.5.28 (S)			X
11	HALF-HOUR CHARGE TEST		X		3.7 (G), 3.8.17 (S)	4.5.29.1 (S)			
12	ONE-HOUR CHARGE TEST			X	3.7 (G), 3.8.17 (S)	4.5.29.2 (S)			
13	CYCLING TEST	X			3.8.9 (G), reqmt 1 of table III (S)	4.5.10 (G)		X	
14	CAPACITY DISCHARGE AT 49°C (120°F)		X		3.8.6 (S), reqmt 2 of table III (S)	4.5.11 (S)			
15	DISCHARGE WHILE INVERTED		X		3.5.9 (G), reqmt 5 of table III (G)	4.5.13 (G)			X
16	ALTITUDE TEST		X		3.5.9 (G), 3.7 (G), 3.8.10 (G)	4.5.14 (G)			X
17	SHOCK TEST (BASIC DESIGN)				3.7 (G), 3.8.10 (G)	4.5.15 (S)		X	
18	TEMPERATURE SHOCK TEST	X			3.7 (G), 3.8.10 (G)	4.5.16 (G)			

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TABLE I. Qualification inspection and conformance inspection of batteries - Continued.

Test number	Examinations and tests 1/	Qualification inspection					Conformance inspection		
		Sample number			Requirement Paragraph 5/	Test Paragraph 5/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
19	TEMPERATURE RISE AND FLOAT TEST		X		3.7 (G), 3.8.11 (G), reqmt 1 of table III (S)	4.5.17 (G)			X
20	BATTERY GAS EMISSION TEST		X		3.8.10 (G), 3.8.12 (G)	4.5.18 (G)			
21	VIBRATION TEST		X		Reqmt 4 (N), 3.7(G), 3.8.10 (G)	4.5.19 (S)			
22	HUMIDITY TEST		X		3.7 (G), 3.8.10 (G)	4.5.20 (G)			
23	SALT FOG TEST	X			3.7 (G), 3.8.10 (G)	4.5.21 (G)			
24	GROUND STORAGE TEST			X 2/	3.8.13.1 (S), 3.8.18 (N)	4.5.22 (S)			
25	DEEP DISCHARGE RECOVERY TEST			X	3.7 (G), 3.8.13.3 (G)	4.5.27 (G)			
26	PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST	X			3.7 (G), 3.8.10 (G), reqmt 1 of table III (S)	4.5.23 (S)			
27	SHOCK TEST (CRASH SAFETY)	X			3.8.10 (G), 3.8.15 (G)	4.5.25 (G)			
28	SHELF LIFE TEST			X	3.8.9 (G), 3.8.13.2 (G)	4.5.26 (G)			
29	IMPACT RESISTANCE TEST (NONMETALLIC BATTERY CONTAINER)		4/		3.8.14 (G)	4.5.24 (G)			
30	FINAL EXAMINATION	X	X	X	3.7 (G), 3.8.4 (G)	4.5.31 (G)		X	X

1/ Except for inspections 3 through 5 (which may be interchanged), inspections shall be performed sequentially in the order listed.
2/ See sample information in test paragraph.
3/ The manufacturer requesting qualification to this specification may submit an additional (fourth) test sample to expedite the qualification test program.
4/ For nonmetallic battery container designs, two outer nonmetallic battery containers shall be furnished for inspection.
5/ The notations in parentheses specify the source of the paragraph. "(G)" means the paragraph as shown in MIL-PRF-8565. "(S)" shows that the paragraph has been modified by this specification sheet. "(N)" indicates that a new paragraph or requirement has been added by MIL-PRF-8565/7.

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7.11. Referenced documents. In addition to MIL-PRF-8565, this specification sheet references MIL-PRF-18148/3.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 170
DLA - CC

Preparing activity:
Navy - AS

(Project 6140-2021-022)

Review activity:
Air Force - 70

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